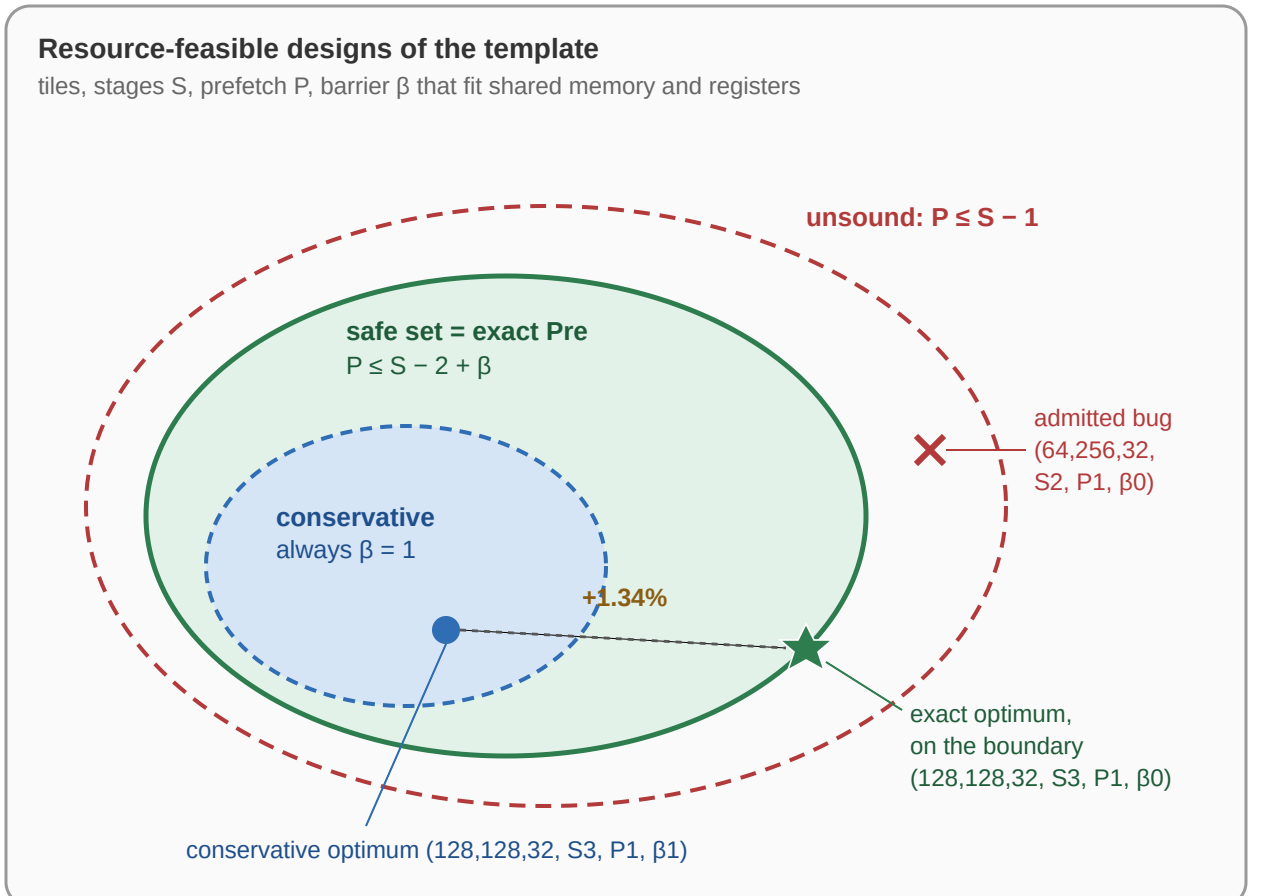


Weakest precondition = least design loss

The template's search constraints imply its precondition (K1), so the solver can only reach designs the precondition admits.

Exact (P ≤ S − 2 + β)

feasible set = safe set; matches the checker on 24/24 schedule cases. MILP optimal in 0.03–0.16 s; each choice certified in ~25 ms; no correctness or design loss.

Too strong (always keep the trailing barrier)

still correct, but excludes safe designs: precondition loss Δpre = +1.34% latency on 1000^3, where barriers matter; 0% on the two memory-bound workloads.

Too weak (P ≤ S − 1, barrier ignored)

admits unsafe designs; on all three workloads the MILP picks one (predicted as fast, less memory). The gate rejects it with a replayed race (Fig. 4a).